Behavioral and Neural Interactions Between Working Memory and Stop-Signal Inhibition

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Introduction

Goal-directed behavior requires the cooperative recruitment of multiple cognitive control processes, like working memory and response inhibition.

Taxonomies of cognitive control processes suggest that working memory and response inhibition are separable[1], but previous studies have not focused on the cooperative recruitment required in everyday behavior.

To interrogate the relationship between working memory (i.e., updating active memory traces and resisting proactive interference from recent memory traces) and response inhibition, we combined N-back[2] and a stop-signal tasks[3].

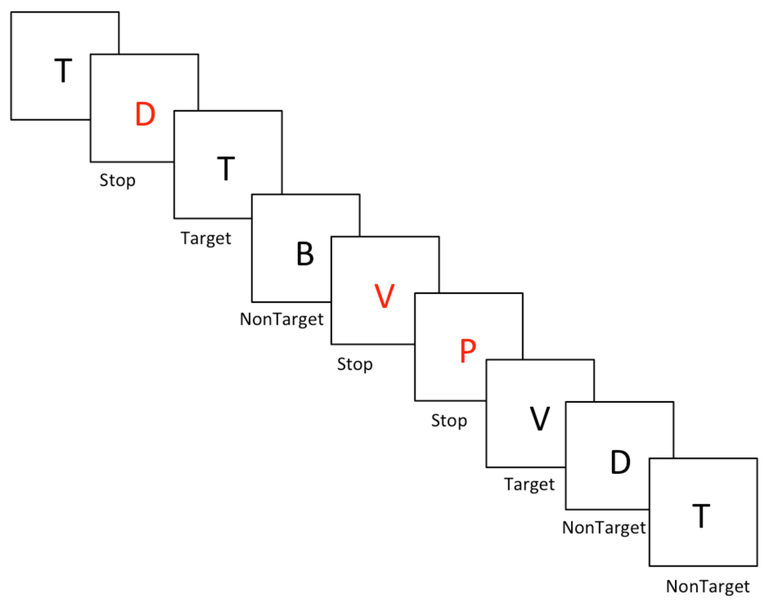
Method/Analyses

N=29 subjects underwent a single 90-minute MRI scanning session (GE 3T scanner) using a multiband acquisition sequence (MB=8; TR=680ms; voxel size=2.2mm; 64 axial slices). All subjects were run through fMRI quality assurance (github.com/poldracklab/mriqc) and preprocessing (github.com/poldracklab/fmriprep) pipelines developed by the Stanford University Center for Reproducible Neuroscience.

Data were smoothed with a kernel with FWHM=4.4mm. First level analysis included task regressors, their temporal derivatives, 12 motion parameters, 6 aCompCor principal components[4], framewise displacement, and stdDVARs. Second-level analyses used ordinary least squares estimation. To threshold the map, we used non-parametric cluster-extent thresholding with the cluster forming threshold u=3.1 (p<0.001) using FSL's randomise.

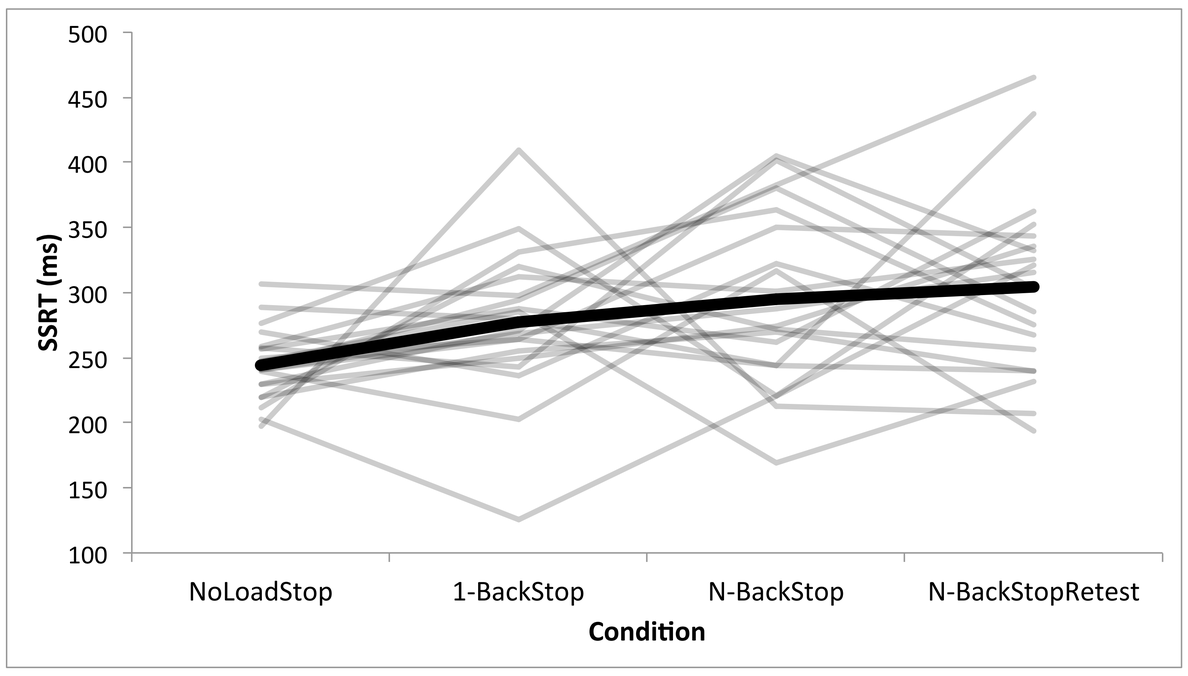
N=20 agreed to return for a follow-up behavioral study that included four tasks: a retest (FollowUp), a 1-back stopping task (1-BackStop), a no working memory load stopping task (NoLoadStop), and an adaptive N-back task with no stop signals (N-BackNoStop).

Task: Example 2-back block in N-BackStop task. N-back adjusted up and down based upon go error rate. If the current letter matched the stimulus presented N trials ago, subjects responded with one button press (target trials, 33% of all trials) but otherwise responded with a different button press (nontarget trials, 67% of all trials). On 1/3 of all trials, the letter turned red after a stop delay, which acted as a stop signal, so subjects were instructed to make neither button press. Subjects were instructed to remember all letters in the sequence even if they turned red.

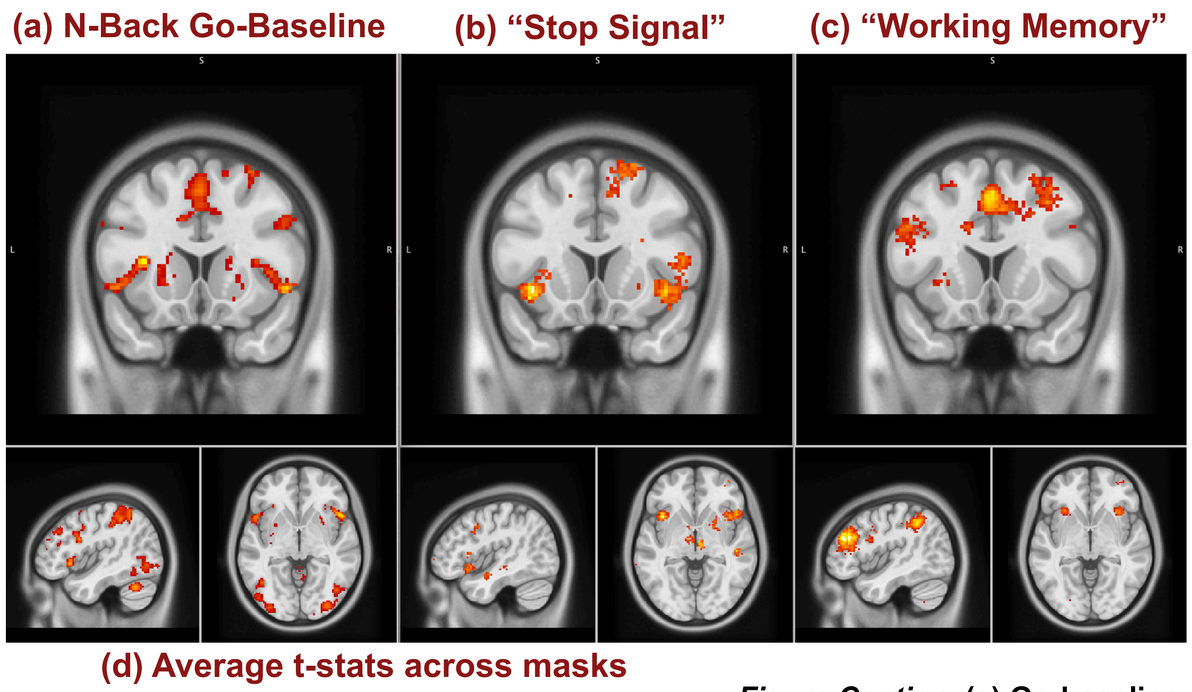


Results

Stop-signal reaction time (SSRT), a measure of response inhibition, increased with N-back load, a measure of working memory demands (ANOVA p < .01)



BOLD Activation (a) overlapped with meta-analysis of stop signal (b) and working memory (c) maps



(d) Average t-stats across masks

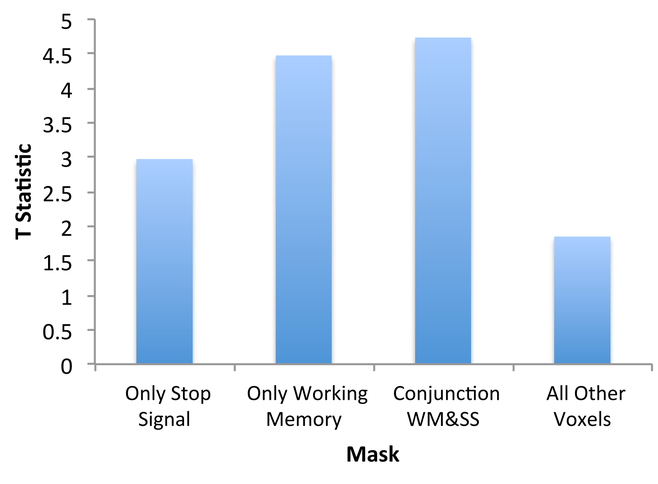


Figure Caption. **(a) Go-baseline contrast from N-BackStop task. (b-c) Neurosynth[5] meta-analytic reverse inference maps for the terms "Stop Signal" and "Working Memory", respectively. (d) Average unthresholded t-statistics from figure a within 4 different masks. Activation was higher in voxels that overlapped with b and c, and especially their conjunction, than all other voxels.**

Negative Results

As presented in our abstract, we pre-registered a hypothesis-driven analysis that looked at current N-back trials that were preceded N-trial-back by stop success or stop failure. In a discovery dataset, we showed slower, less accurate responses when the trial N back was a stop success. We also showed right inferior frontal gyrus and parahippocampal gyrus activity. In a replication dataset and after improving our analysis pipeline, we were unable to reproduce these effects, and the combined analyses of the two datasets showed no effects. We present these negative results as a small step towards mitigating the "file drawer problem".

Conclusions

We showed behavioral evidence for an interaction between N-back and stopping performance, with response inhibition slowing as working memory demands increased. This brings into question existing taxonomies suggesting that response inhibition and working memory are separable.

N-BackStop activation map was similar to a meta-analysis of stop signal and working memory activation, and particularly their conjunction, suggesting that this task successfully recruits the confluence of these processes.

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